Equations of Lines in 3D: Vector Equations

Questions 1–59

Question 1. Find the vector equation of the line that:

a) passes through $(2, -7, 1)$ and is parallel to $3\mathbf{i} + 9\mathbf{j} - 4\mathbf{k}$

b) passes through $(3, 8, -3)$ and is parallel to $-4\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}$

Question 2. Find a vector equation of the line passing through the following pairs of points:

a) $(0, 0, 0)$ and $(4, -1, 3)$

b) $(1, 4, 1)$ and $(3, 5, 7)$

c) $(2, -4, 5)$ and $(3, -1, 6)$

d) $(9, -8, -2)$ and $(9, -3, -8)$

e) $(6, -4, 13)$ and $(1, -4, 8)$

f) $(-1, 10, 3)$ and $(-9, -4, 3)$

Question 3. It is given that l is the line with equation $\mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}$. Given also that points A and B lie on the line l with $\lambda = 2$ and $\lambda = -2$ respectively, find the exact length of the line joining A and B.

Question 4. Show that the point $(6, -4, -12)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} 5 \\ -2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 6 \\ 27 \end{pmatrix}$.

Question 5. Show that the point $(-12, \frac{31}{2}, -\frac{27}{2})$ does not lie on the line with equation $\mathbf{r} = -2\mathbf{i} + 13\mathbf{j} - 6\mathbf{k} + \lambda(4\mathbf{i} - \mathbf{j} + 7\mathbf{k})$.

Question 6. Determine whether the point $(-2, 16, -40)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} 7.5 \\ -3 \\ 2.5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix}$.

Question 7. The point $(5, -10, 2)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} 2 \\ -4 \\ k \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$, where k is a constant. Find the value of k.

Question 8. The point $(-3, -83, -98)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} 4 \\ -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} k \\ 4 \\ 5 \end{pmatrix}$, where k is a constant. Find the value of k.

Question 9. Determine whether the points $A(1, -1, -5)$, $B(-5, -9, -1)$ and $C(16, 19, -15)$ are collinear.

Question 10. Determine whether the points $A(40, -46, -18)$, $B(-8, 18, -10)$ and $C(4, 2, -11)$ are collinear.

Question 11. It is given that $A(7, p, p)$, $B(-1, 11, 27)$ and $C(17, -16, q)$, where p and q are constants, are collinear. Find the value of p and the value of q.

Question 12. Determine whether the following pair of lines are coincident, parallel or neither:

$$l_1 : \mathbf{r} = \begin{pmatrix} 5 \\ -2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 8 \\ 13.5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 7 \\ 8 \\ -9 \end{pmatrix} + \lambda \begin{pmatrix} 0.5 \\ -2 \\ -3.375 \end{pmatrix}.$$

Question 13. Find the coordinates of the point of intersection of the two lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 6 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ -5 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 2 \\ 9 \end{pmatrix}.$$

Question 14. Find the coordinates of the point of intersection of the two lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 4 \\ -1 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 2 \\ 10 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 6 \\ -2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 8 \\ -1 \\ 8 \end{pmatrix}.$$

Question 15. Determine whether the following pairs of lines are coincident, parallel, skew or if they intersect. If they intersect, find the coordinates of the point of intersection:

a) $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 9 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -9 \\ 16 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} -4 \\ 3 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} -4 \\ 6 \\ -\frac{32}{3} \end{pmatrix}$

b) $l_1 : \mathbf{r} = \begin{pmatrix} -5 \\ -2 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -8 \\ -5 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ -7 \end{pmatrix} + \mu \begin{pmatrix} -4.5 \\ 11.5 \\ -7.5 \end{pmatrix}$

c) $l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ -9 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ -8 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 5 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 6 \\ 7 \end{pmatrix}$

Question 16. It is given that the following lines intersect:

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ -4 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ k \\ 8 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} -3 \\ 5 \\ -9 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -6 \\ 4 \end{pmatrix},$$

where k is a constant.

a) Find the value of k.

b) Find the position vector of the point of intersection of the two lines.

Question 17. It is given that the following lines are perpendicular:

$$l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -5 \\ 3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 7 \\ -3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 6 \\ k \\ -1 \end{pmatrix},$$

where k is a constant. Find the value of k.

Question 18. It is given that the line l_1 has the equation $\mathbf{r} = \begin{pmatrix} 1 \\ 9 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$, and that the line l_2 is parallel to the vector $24\mathbf{i} + 4\mathbf{j} - 9\mathbf{k}$. Show that l_1 and l_2 are perpendicular.

Question 19. It is given that the line l_1 has the equation $\mathbf{r} = \mathbf{i} + \lambda(2\mathbf{i} - \mathbf{j} - \mathbf{k})$, and that the line l_2 has the equation $\mathbf{r} = 3\mathbf{j} + 4\mathbf{k} + \mu(\mathbf{j} - \mathbf{k})$. Determine whether l_1 and l_2 are perpendicular.

Question 20.

a) Show that $-2\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$ and $3\mathbf{i} - 5\mathbf{j} - \mathbf{k}$ are not perpendicular vectors.

b) Find the coordinates of the point of intersection of the two lines $l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 6 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} -13 \\ 28 \\ 24 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix}$.

c) It is given that the line l_3 intersects both l_1 and l_2, and is perpendicular to both l_1 and l_2. Find the equation of the line l_3.

Question 21. Find the acute angle between each pair of lines:

a) $l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -7 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ -5 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$

b) $l_1 : \mathbf{r} = \begin{pmatrix} -6 \\ 3 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 5 \\ 2 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 7 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 0 \\ 3 \end{pmatrix}$

c) $l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 0 \\ -10 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ -8 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} 7 \\ 4 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} -7 \\ 6 \\ 6 \end{pmatrix}$

d) $l_1 : \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -9 \\ 2 \end{pmatrix}$, $l_2 : \mathbf{r} = \begin{pmatrix} -11 \\ 8 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} -15 \\ 3 \\ 1 \end{pmatrix}$

Question 22. Find the acute angle between the line $\mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -7 \\ 6 \end{pmatrix}$ and the:

a) x-axis
b) y-axis
c) z-axis

Question 23. Find the obtuse angle between the following pair of lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -12 \\ 3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ -5 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 10 \\ 1 \\ 5 \end{pmatrix}.$$

Question 24. Find the obtuse angle between the following pair of lines:

$$l_1 : \mathbf{r} = 2\mathbf{i} + 5\mathbf{j} - \mathbf{k} + \lambda(\mathbf{i} - \mathbf{j} + \mathbf{k}), \quad l_2 : \mathbf{r} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k} + \mu(2\mathbf{i} + 4\mathbf{j} + 11\mathbf{k}).$$

Question 25. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 6 \\ 2 \end{pmatrix}$. l_1 is mapped to its image l_2 by the matrix $\begin{pmatrix} -1 & 2 & 3 \\ 1 & -1 & 2 \\ 0 & 2 & 1 \end{pmatrix}$. Find the equation of the line l_2.

Question 26. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 7 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 5 \\ -6 \end{pmatrix}$. l_1 is mapped to its image l_2 by each of the following transformations. Find the equation of the line l_2 in each case:

a) Stretch, parallel to the x-axis, scale factor 2
b) Enlargement, centred at O, scale factor 3
c) Reflection in the plane $x = 0$
d) Reflection in the plane $y = 0$
e) Rotation, anticlockwise about the y-axis, by 90°
f) Rotation, anticlockwise about the z-axis, by 90°

Question 27. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}$. The line l_1 is mapped to its image l_2 by the transformation associated with each of the following matrices. Find the acute angle between l_1 and l_2 in each case:

a) $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 1 & 1 \end{pmatrix}$

b) $\begin{pmatrix} -2 & 3 & 1 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 5 & -3 & -4 \\ 2 & 8 & -3 \\ -5 & -4 & -1 \end{pmatrix}$

Question 28. Write down the coordinates of a general point, in terms of the given parameter, that lies on each of the following lines:

a) $\mathbf{r} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$

b) $\mathbf{r} = \begin{pmatrix} 5 \\ -9 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ -3 \\ 16 \end{pmatrix}$

Question 29. The line l has equation $\mathbf{r} = \begin{pmatrix} -4 \\ 2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 8 \\ -4 \end{pmatrix}$. State the coordinates of the point on the line with the given value of λ:

a) $\lambda = 1$
b) $\lambda = -1$
c) $\lambda = -4$

Question 30. It is given that the point A has coordinates $(2, 1, -5)$. The point B lies on the line l which has the equation $\mathbf{r} = \begin{pmatrix} 2 \\ 2 \\ -7 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$. Find $\overrightarrow{AB}$ in terms of λ.

Question 31. It is given that the point P has coordinates $(3, -7, 2)$. The point Q lies on the line l which has the equation $\mathbf{r} = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -2 \\ 5 \end{pmatrix}$. Find $\overrightarrow{QP}$ in terms of λ.

Question 32. Find the value of λ in each case such that the vector $\begin{pmatrix} 2-\lambda \\ 1+5\lambda \\ 2\lambda - 1 \end{pmatrix}$ is perpendicular to:

a) $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

b) $\begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}$

c) $\begin{pmatrix} -1 \\ 3 \\ -5 \end{pmatrix}$

Question 33. The line l_1 passes through the points A and B with coordinates $(3, 7, -1)$ and $(2, 5, -3)$ respectively. The line l_2 has equation $\mathbf{r} = \begin{pmatrix} 2 \\ -5 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 6 \\ -2 \\ -3 \end{pmatrix}$.

a) Find a vector equation of the line l_1.

b) Show that $P(38, -17, -15)$ lies on l_2. It is given that C lies on l_1 such that CP is perpendicular to l_2.

c) Find the coordinates of C.

Question 34. The points A and B have coordinates $(2, -1, 2)$ and $(3, 8, -2)$ respectively. The point C is such that $\overrightarrow{OC} = 99\overrightarrow{OA}$. The line l is parallel to AB and passes through C.

a) Find a vector equation of the line l. The point D lies on l such that angle $ABD = 90°$.

b) Find the coordinates of D.

Question 35. Two lines are given by

$$l_1 : \mathbf{r} = \begin{pmatrix} 10.25 \\ -6 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 4 \\ 0 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} a \\ 1 \\ 7 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ b \end{pmatrix},$$

where a and b are positive constants. It is also given that the lines l_1 and l_2 intersect and form an angle of $120°$. Find the exact values of the constants a and b.

Question 36. The points A and B have position vectors $2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $3\mathbf{i} - \mathbf{j} - 4\mathbf{k}$ relative to a fixed origin O. The points A and B lie on the line l.

a) Find a vector equation for the line l.

The point C has position vector $a\mathbf{i} + \mathbf{j} - \mathbf{k}$, where a is a constant. It is given that AC is perpendicular to the line l.

a) Find the value of a.

b) Find the area of the triangle ACB.

Question 37. It is given that the line l has equation $\mathbf{r} = \begin{pmatrix} 3 \\ -5 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 8 \\ -9 \end{pmatrix}$.

a) The point A lies on the line l when $\lambda = -2$. Find the coordinates of A.

b) Write down the vector equation of a line that is perpendicular to l and passes through A.

Question 38. It is given that:

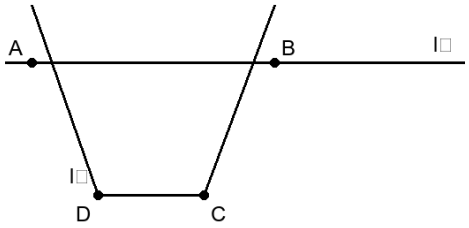
$$l_1 : \mathbf{r} = \begin{pmatrix} -7 \\ 3 \\ 10 \end{pmatrix} + \alpha \begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} + \beta \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix},$$

$$l_3 : \mathbf{r} = \begin{pmatrix} -21 \\ 5 \\ 23 \end{pmatrix} + \gamma \begin{pmatrix} 4 \\ -1 \\ -5 \end{pmatrix}, \quad l_4 : \mathbf{r} = \begin{pmatrix} 19 \\ 1 \\ -9 \end{pmatrix} + \delta \begin{pmatrix} -6 \\ 0 \\ 3 \end{pmatrix}.$$

a) Show that the points where these four lines intersect form a parallelogram but not a rhombus.

b) Hence find the exact area of the parallelogram.

Question 39. The diagram below shows the symmetrical trapezium $ABCD$, where $|AD| = |BC|$.



The points A and B have coordinates $(6, -6, 1)$ and $(14, -12, 5)$ respectively. A and B lie on the line l_1.

a) Find a vector equation of the line l_1.

It is given that the line l_2 passes through A and D and has equation $\mathbf{r} = \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$.

a) Show that l_1 and l_2 intersect at A.

b) D lies on l_2 when $\mu = 4$. Find the coordinates of D.

c) Find the coordinates of C.

Question 40. It is given that A, B, C and D have coordinates $(4, 7, -6)$, $(1, 4, -6)$, $(3, 2, -9)$ and $(3, 3, -9)$ respectively. A and B lie on the line l_1, while C and D lie on the line l_2. l_1 and l_2 intersect at the point P.

a) Find the coordinates of P.

b) E lies on l_2 such that $|BP| = |PE|$. Find the possible coordinates of E.

Question 41. A, B, C and D have coordinates $(4, -5, 4-\sqrt{6})$, $(4+\sqrt{6}, -5-3\sqrt{6}, -2-\sqrt{6})$, $(3+\sqrt{6}, 6, 1)$ and $(\sqrt{6}-9, 1-6\sqrt{6}, 5-2\sqrt{6})$ respectively. A and B lie on the line l_1, while C and D lie on the line l_2. A and C lie on a sphere, whose centre is the point of intersection of l_1 and l_2. Find the volume of the sphere.

Question 42. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -3 \\ 5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 5 \\ -8 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} 16 \\ -8 \\ -23 \end{pmatrix}.$$

The lines l_1 and l_2 intersect at the point P.

a) Find the coordinates of P. The point A lies on l_1 such that $|AP| = 849\sqrt{43}$. The point B lies on l_2 such that angle $ABP = 90°$.

b) Find the two possible coordinates of B.

Question 43. A, B and C have coordinates $(3, 4, -5)$, $(-1, 3, -2)$ and $(3, p, q)$ respectively, where p and q are constants and $p > 0$. A and B lie on the line l.

a) Find the coordinates of C, given that C is $\sqrt{90}$ units from A and that angle $CAB = 90°$. The point D lies on the line l such that the triangle CAD has twice the area of the triangle CAB.

b) Find the possible coordinates of D.

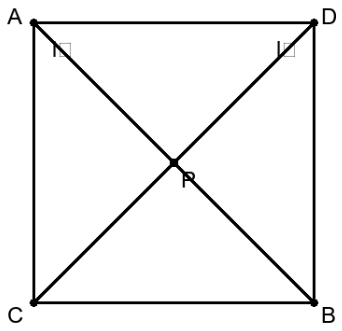
Question 44. It is given that the line l_1 passes through the points A and B with coordinates $(-1 + 3\sqrt{89}, 5 - 2\sqrt{89}, 9 + 6\sqrt{89})$ and $(-1 - 3\sqrt{89}, 5 + 2\sqrt{89}, 9 - 6\sqrt{89})$ respectively.

a) State the midpoint of A and B.

The line l_2 has equation $\mathbf{r} = \begin{pmatrix} 3 \\ 23 \\ 13 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 9 \\ 2 \end{pmatrix}$.

a) l_1 and l_2 intersect at the point P. Find the coordinates of P.

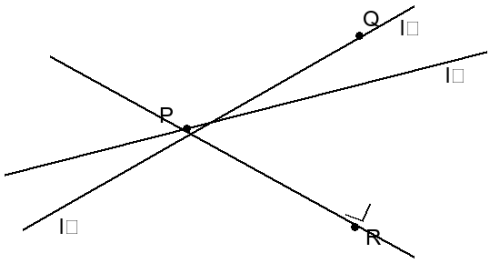
b) The points C and D lie on l_2 such that $ADBC$ is a square, as shown in the diagram.



Find the coordinates of C and D.

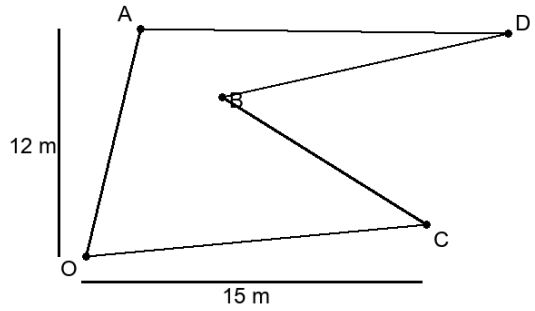
Question 45. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} 5 \\ -3 \\ -9 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 7 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -6 \\ 1 \\ 2 \end{pmatrix}.$$



The point P has coordinates $(18, 4, -1)$ and is the point where l_2 and l_3 intersect. The line l_3 is parallel to l_1. Q lies on the line l_3. R lies on the line l_2 when $\mu = -27$. It is given that QR is perpendicular to l_2. Find the coordinates of Q.

Question 46. A drone light show involves two drones that are programmed to follow straight paths at different angles. The drones are programmed to remain within the symmetrical trapezium shown in the diagram, as seen from above.



Take the point O to have coordinates $(0, 0, 0)$. Drone 1 starts from 1 metre above point A and follows a straight path to reach a height of 15 metres above point C. Meanwhile, Drone 2 starts from 4 metres above point B and follows a straight path to reach a height of k metres above point D.

a) Given that the drones' flight paths intersect one another, find the value of k.

It is given that the drones are programmed to set off at the same time and both fly at constant speeds. It takes Drone 1 five seconds to fly from its initial position 1 metre above A to its final position 15 metres above C.

a) How fast would Drone 2 need to fly to crash into Drone 1?

b) Describe two assumptions you have made in answering part (b).

Question 47. The points A and B have coordinates $(6, 2, -2)$ and $(9, 3, -3)$ respectively.

a) Show that the vector equation of the line passing through A and B can be given by $\mathbf{r} = \lambda \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$.

The points A and B are transformed to their images C and D respectively by a rotation, anticlockwise about the z-axis by $90°$.

a) Find the equation of the line passing through C and D.

b) Find the acute angle between the line in part (a) and the line found in part (b).

Question 48. It is given that the line l_1 has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$. l_1 is mapped to its image l_2 by the matrix $\begin{pmatrix} 1 & 2 & 0 \\ -1 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$. Find the equation of the line l_2.

Question 49. It is given that the line l_1 has equation $\mathbf{r} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k} + \lambda(3\mathbf{i} - \mathbf{j} - \mathbf{k})$. l_1 is mapped to its image l_2 by the matrix $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix}$. Find the equation of the line l_2.

Question 50. It is given that $l_1 : \mathbf{r} = \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 10 \\ 6 \\ -11 \end{pmatrix}$. Show that the line l_1 can be mapped to the line l_2 by the transformation represented by the matrix $\begin{pmatrix} 1 & -2 & 1 \\ 1.5 & -1 & 0.25 \\ 1 & 0.5 & -3 \end{pmatrix}$.

Question 51. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} 8 \\ 2 \\ -5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 26 \\ -15 \\ -10 \end{pmatrix} + \mu \begin{pmatrix} a \\ 16 \\ b \end{pmatrix}, \quad A = \begin{pmatrix} p & 1 & -3.5 \\ 3 & -3 & 2 \\ 2.5 & -1 & q \end{pmatrix}.$$

Given also that $\det A \neq 0$, l_1 can be mapped to l_2 by the transformation represented by matrix A. Find the values of the constants a, b, p and q.

Question 52. It is given that the line l has equation $\mathbf{r} = \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}$. l intersects a circle centred at $(1, 1, 0)$ with radius $\sqrt{14}$ at two distinct points. Find the coordinates of these two points.

Question 53. The plane Π has equation $3x - 2y + 5z = 6$. The line l has equation $\mathbf{r} = \begin{pmatrix} 40 \\ 7 \\ -20 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 9 \\ 6 \end{pmatrix}$. Show that l lies in the plane Π.

Question 54. The plane Π contains the following two lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 9 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ -1 \\ -4 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -3 \\ -8 \end{pmatrix}.$$

Find a vector equation of the plane Π.

Question 55. The plane Π contains the following two lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ -6 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 6 \\ -3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 3 \\ 25 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 9 \\ 18 \end{pmatrix}.$$

Find a cartesian equation for the plane Π.

Question 56. The plane Π contains the following two lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 6 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 9 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ k \\ -5 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ 5 \\ 6 \end{pmatrix}.$$

Find the value of the constant k.

Question 57. The plane Π contains the following two lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} k \\ 3 \\ 1 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ -2 \\ 8 \end{pmatrix},$$

where k is a constant. Find a cartesian equation of the plane Π that is independent of k.

Question 58. Write the following vector equations of lines in cartesian form:

a) $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 5 \\ -7 \end{pmatrix}$

b) $\mathbf{r} = \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$

c) $\mathbf{r} = \begin{pmatrix} 4 \\ -8 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 6 \end{pmatrix}$

d) $\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ -8 \end{pmatrix}$

e) $\mathbf{r} = \begin{pmatrix} 7 \\ -9 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -5 \\ 0 \end{pmatrix}$

f) $\mathbf{r} = \begin{pmatrix} -2 \\ 1 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

Question 59. Write the following vector equations of lines in vector product form:

a) $\mathbf{r} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$

b) $\mathbf{r} = \begin{pmatrix} 0 \\ -6 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -2 \\ -5 \end{pmatrix}$